

Most of the Pre-College program focuses on a specific area of computational biology: specifically, the genomic study of the DNA of microbes and viruses. We aim to show you the immense array of deep questions that we can form from just one area of study, and the beauty of the algorithms for answering these questions computationally.

But we also want you to experience the much greater breadth of topics in modern biology that touch on computation. And we don't just mean learn about them; we mean *build* them! To cap off the summer, you will work with your project group to explore and implement a technology in computational biology. By the end, each group will have gained both theoretical and hands-on knowledge in their chosen project area that they will share with their peers.

Project Overview

This project will have two main parts:

- **Part 1: Review of the topic**
Research your topic and present an overview of its history, its significance, and how it works. Make sure to include any necessary biological and computational or mathematical context.
- **Part 2: Algorithm application**
Implement an algorithm related to your topic on an available dataset and build a website visualizing the result. You may leverage LLMs for this part, just make sure you know what your code is doing!

Throughout the process, you will submit three checkpoint deliverables:

- **Checkpoint 0: Topic choice** (due W. June 24)
Choose a topic from the list below and submit your choice on Canvas. You can use the example paper as a basis for your project and/or your research on the topic.
- **Checkpoint 1: Research check-in** (due W. July 1)
Compose a one-page progress report containing the following (submit on Canvas):
 - A one-paragraph intro to your topic
 - At least 4 reputable references
 - A short description of what each group member contributed to the project up to this checkpoint
 - A description of any major blocks or challenges that your group may need help with (focus on the research and understanding the topic)

Checkpoint 2: Code check-in (due W. July 8)

- Compose a one-page progress report containing the following (submit on Canvas):
- An outline of how you plan to apply or extend a model using publicly available data, including any draft code you may have
 - A short description of what each group member contributed to the project up to this checkpoint
 - A description of any major blocks or challenges that your group may need help with (focus on the code/application)

At the end of the program, you will submit your **final project deliverable in the form of a website**. Slides/videos to supplement are optional but encouraged where they may help your audience understand your work.

Your project will culminate in a 10-minute presentation at the end of the summer, covering both your research on the topic and a demo showing off your code's results.

Your audience for this presentation is the other students in the program, so you should plan your talk accordingly. Try to make your presentation as fun and educational as possible, and feel free to be creative!

On the penultimate day in our program, you will have the opportunity to present your work to your peers, and the best presentations (voted for by students) will be invited to present to a larger audience on the final day of the program.

Project Structure and Schedule

This project effectively has four parts:

1. **Research:** Read about your topic and ensure you understand it well enough to teach your peers.
This will be the focus of the first week of the program.
2. **Build:** Start planning and implementing code to apply methodologies in your topic area to new data.
This will be the focus of the second-third week of the program.
3. **Apply:** Gather publicly available data to use apply your code to and get interesting results.
This will be the focus of the second-third week of the program.
4. **Communicate:** Create and present a multimedia website to teach your peers about your topic and demonstrate your work.
This will be the focus of the final week of the program.

Notably, these are roughly the components of scientific research in computational biology!

Each section will have specific, structured guidance during its designated weeks, but you are encouraged to think about all four parts throughout the program.

There will be one afternoon project work session per week for each group, during which we will check in about progress, help guide ideation and implementation, and answer questions. To make the most of these work sessions, be sure to come prepared with an idea of what has worked, what hasn't worked, and what your group needs help to understand.

Remember that we are here to help, and science is about asking questions!

Project topics

Biology has a huge variety of topics that we won't directly cover in this course. Below is a list of potential topics you can choose from. The final option, "Student's choice", is left open in case your group would like to choose a topic not on this list.

Each topic choice includes a foundational paper in the field ("Ref") and a short note describing the topic of this paper. These sources can provide a good starting point for your research and give you ideas on how to implement the code portion of your project. We will discuss ways to dive deeper into your topic and find additional sources during project work sessions.

Topic choice assignment: Submit three numbers between 1 and 29 indicating your top three choices. For example, if you submit "23, 29 (insert project topic), 3", then we assume that project 23 is your first choice, project 29 is your second choice, and project 3 is your third choice. If you choose 29, you should also provide your topic.

1. How do we predict the 3-D shape of a protein from a gene sequence?
Ref: Šali, A., & Blundell, T. L. (1993). Comparative protein modelling by satisfaction of spatial restraints. *Journal of Molecular Biology*, 234(3), 779–815.
<https://doi.org/10.1006/jmbi.1993.1626>
Note: This paper introduces a method to find the most probable 3D structure of a protein using comparisons with known structures.
2. How can we structural predict properties of a protein from an amino acid sequence?
Ref: Chou, P. Y., & Fasman, G. D. (1974). Prediction of protein conformation. *Biochemistry*, 13(2), 222–245. <https://doi.org/10.1021/bi00699a002>
Note: This paper focuses on modeling and prediction of secondary structure from empirical data.
3. How are similar protein sequences found for use in methods like homology modeling?
Ref: Altschul et al. (1990). Basic local alignment search tool. *Journal of Molecular Biology*, 215(3), 403–410. [https://doi.org/10.1016/s0022-2836\(05\)80360-2](https://doi.org/10.1016/s0022-2836(05)80360-2)
Note: This is the original BLAST paper, mentioned in Hackathon lecture.
4. How can genes be edited to cure diseases?
Ref: Doudna & Charpentier (2012). A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science*, 337(6096), 816–821.
<https://doi.org/10.1126/science.1225829>
Note: This is the original CRISPR paper, a common topic that comes up in lab.
5. How can we computationally predict RNA structure for use with methods like CRISPR?
Ref: Zuker, M., & Steigler, P. (1981). Optimal computer folding of large RNA sequences using thermodynamics and auxiliary information. *Nucleic Acids Research*, 9(1), 133–148. <https://doi.org/10.1093/nar/9.1.133>
Note: This paper introduces a methodology to find the minimal-energy (most stable)

conformation of an RNA molecule, a useful method in RNA design for applications like CRISPR.

6. What is a neural network, and how has biology informed the construction of neural networks?

Ref: McCulloch & Pitts (1943). A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*, 5, 115–133.

<https://www.cs.cmu.edu/~epxing/Class/10715/reading/McCulloch.and.Pitts.pdf>
Note:

7. What algorithms have researchers borrowed from nature in order to solve problems inside and outside of biology?

Ref: Kirkpatrick, Gelatt & Vecchi (1983). Optimization by simulated annealing. *Science*, 220(4598), 671–680. <https://doi.org/10.1126/science.220.4598.671>

Note: This paper introduces a commonly used algorithm based on the behavior of metals in response to temperature changes.

8. How can we train a computer to find organelles within microscopic images of cells?

Ref: Otsu (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62–66.

<https://doi.org/10.1109/TSMC.1979.4310076>

Note: This paper introduces a commonly used image thresholding algorithm in biology.

9. How are resources like Ancestry, 23AndMe, etc. able to determine the breakdown of someone's ethnicity from their saliva?

Ref: Rosenberg et al. (2002). Genetic structure of human populations. *Science*, 298(5602), 2381–2385. <https://doi.org/10.1126/science.1078311>

Note: This paper uses clustering algorithms to identify genetic similarities between different groups of humans. Data used comes from the [HGDP-CEPH Human Genome Diversity Cell Line Panel](#).

10. How do we compare multiple sequences for use in tools like AlphaFold?

Ref: Thompson, J. D., Higgins, D. G., & Gibson, T. J. (1994). CLUSTAL W: improving the sensitivity and utility of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucleic Acids Research*, 22(22), 4673–4680. <https://doi.org/10.1093/nar/22.22.4673>

Note: This paper introduces an algorithm for multiple sequence alignment that builds on some foundational methods covered in hackathon.

11. Why is sequencing a cancer tumor useful for prescribing therapeutics to the patient?

Ref: Alexandrov et al. (2013). Signatures of mutational processes in human cancer. *Nature*, 500(7463), 415–421. <https://doi.org/10.1038/nature12477>

Note: This paper discusses the utility of analyzing and understanding mutation signatures in different tumors for cancer treatment and prevention.

12. How can we use metagenomics to study the human gut microbiome?

Ref: Steven R. Gill et al. (2006). Metagenomic Analysis of the Human Distal Gut Microbiome. *Science*, 312, 1355–1359. <https://doi.org/10.1126/science.1124234>

Note: This paper presents a view of the gut microbiome's function and compares it

to human metabolic processes to build a picture of human metabolism as a synergistic process. Its methods focus on use of databases and literature to annotate new data.

13. How do we measure the levels at which genes are turned on and off and use this to compare species as well as healthy and diseased individuals?

Ref: Schena et al. (1995). Quantitative monitoring of gene expression patterns with a complementary DNA microarray. *Science*, 270(5235), 467–470.

<https://doi.org/10.1126/science.270.5235.467>

Note: This paper introduces an early system of measuring gene expression from which we can obtain data for computational analysis.

14. How can we use robotically operated biological laboratories to solve problems faster and more accurately?

Ref: Laverick, A, et al. (2025) OT-Mation: an open-source code for parsing CSV files into Python scripts for control of OT-2 liquid-handling robotics. *Synthetic Biology*, 10(1). <https://doi.org/10.1093/synbio/ysaf009>

Note: This paper introduces a Python interface for automated biology, with application examples demonstrating the vast possibilities of this technology. Ask Josh if you're interested in learning more :)

15. How can we use single-cell RNA-seq data and clustering algorithms to differentiate cell types and improve our understanding of tissues?

Ref: Macosko et al. (2015). Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets. *Cell*, 161(5), 1202–1214.

<https://doi.org/10.1016/j.cell.2015.05.002>

Note: This paper introduces a wet lab method for high-throughput single-cell transcriptomics and mentions some clustering methods used for analysis.

16. How can we use spatial transcriptomics to reconstruct a 3-D "map" of gene activity in intact tissues?

Ref: Ståhl et al. (2016). Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science*, 353(6294), 78–82.

<https://doi.org/10.1126/science.aaf2403>

Note: This paper introduces spatial transcriptomics through application on mouse brain and breast cancer tissues.

17. How can we understand the function of non-coding DNA variants to predict whether they may cause disease?

Ref: ENCODE Project Consortium (2012). An integrated encyclopedia of DNA elements in the human genome. *Nature*, 489, 57–74.

<https://doi.org/10.1038/nature11247>

Note: This paper introduces a database of noncoding regions of the human genome, along with applications to identify their functions/effects in the body or in disease.

18. How can we model the transmission of disease (such as COVID-19) and determine how to limit or prevent spread?

Ref: Kermack, W. O., & McKendrick, A. G. (1927). A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London A*,

115(772), 700–721. <https://doi.org/10.1098/rspa.1927.0118>

Note: This early paper introduces the SIR model, a foundational model in epidemiology that is the basis of more complex methods used today.

19. What is machine-learning-guided directed evolution, and how might it help improve human society?

Ref: Cunningham, B. C., & Wells, J. A. (1989). High-resolution epitope mapping of hGH-receptor interactions by alanine-scanning mutagenesis. *Science*, 244(4908), 1081–1085. <https://doi.org/10.1126/science.2471267>

Note: This paper introduces alanine-scanning, an early method that provides the conceptual basis for more complex directed evolution methods.

20. What are protein–protein interaction networks, how are they useful, and what can we learn from them?

Ref: Lewis, A., Saeed, R., & Deane, C. (2009). Predicting protein–protein interactions in the context of protein evolution. *Molecular BioSystems (MBS)*, 6(1), 55–64. <https://doi.org/10.1039/B916371A>

Note: This paper reviews methods for predicting protein-protein interactions, many of which are used in popular interaction databases such as [STRING](#).

21. How can we use simulation of physiological processes *in silico* to support disease research?

Ref: Dalla Man, C., Raimondo, D. M., Rizza, R. A., & Cobelli, C. (2007). GIM, simulation software of meal glucose-insulin model. *Journal of diabetes science and technology*, 1(3), 323–330. <https://doi.org/10.1177/193229680700100303>

Note: This paper introduces a model of glucose and insulin levels to support diabetes research. The math behind the model is laid out clearly, and demonstrates a focused method that could fit into larger-scale simulation studies.

22. How can we identify new potential drugs and computationally estimate their “goodness of fit”?

Ref: Trott, O., & Olson, A. J. (2010). AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2), 455–461. <https://doi.org/10.1002/jcc.21334>

Note: This paper introduces a widely used method to “score” drug candidates against a specific protein associated with a target disease/condition.

23. How can logic-gate gene circuits be designed and used?

Ref: Gardner, Cantor & Collins (2000). Construction of a genetic toggle switch in *Escherichia coli*. *Nature*, 403, 339–342. <https://doi.org/10.1038/35002131>

Note: This paper demonstrates a genetic “switch” in *E. coli* that turns on/off expression of a specific gene.

24. How can wearable technology improve healthcare monitoring, and what computational methods enable this?

Ref: Miao, F., Wang, X., Yin, L. & Li, Y. (2019). A Wearable Sensor for Arterial Stiffness Monitoring Based on Machine Learning Algorithms. *IEEE Sensors Journal*, 19(4), 1426–1434. <https://doi.org/10.1109/JSEN.2018.2880434>

Note: This paper introduces a methodology to monitor arterial stiffness using a healthcare wearable device, using tree algorithms and linear regression as a basis.

25. How can we model the ways that wildlife interact with humans and human made communities?

Ref: Phillips, S. J., Anderson, R. P., & Schapire, R. E. (2006). Maximum entropy modeling of species geographic distributions. *Ecological Modelling*, 190(3–4), 231–259. <https://doi.org/10.1016/j.ecolmodel.2005.03.026>

Note: This paper introduces and applies a model for predicting where animal populations may be found using limited data.

26. How can we measure and compare the microbes that live in different environments?

Ref: Lozupone, C., & Knight, R. (2005). UniFrac: a new phylogenetic method for comparing microbial communities. *Applied and Environmental Microbiology*, 71(12), 8228–8235. <https://doi.org/10.1128/AEM.71.12.8228-8235.2005>

Note: This paper introduces a method for comparison of the microbiome in different samples, a computational problem that would be downstream of your environmental data collection from lab.

27. Student's choice (If you have a topic of your own that you would be excited about working on and that has a clear computational focus, we would be happy to hear your idea!)